OWN YOUR AI #5 · PRACTITIONER SESSION

Local LLMs without being a developer

Copy-pasting diagnoses out of medical reports

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WHO I AM

About me and this talk

Not a developer

Process automation manager.

A side project

Together with a friend who is a doctor.

Built with Claude Code

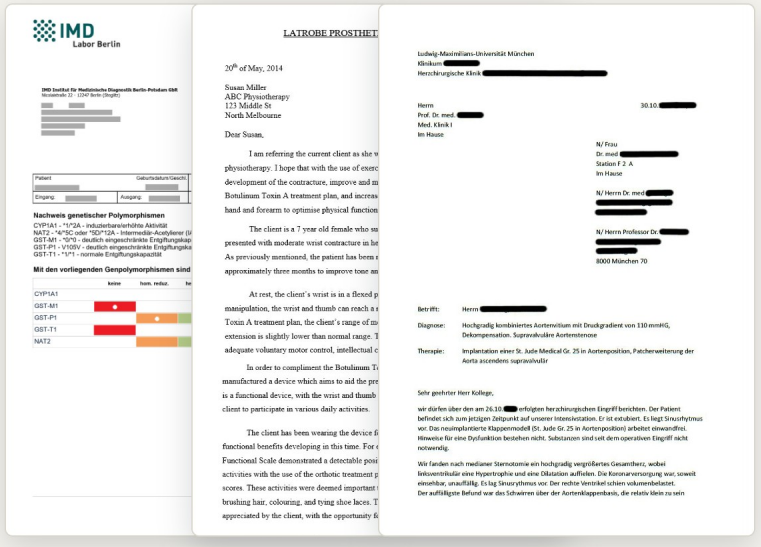
I describe the goal, it writes and runs the code.

What you get today

The non-developer approach. I’m happy about your input.

WHAT THE WORK LOOKS LIKE

From many pages to one list



copy & paste

by hand, document by document



“Why don’t we try that locally, with an LLM?”

Left: publicly available example documents · Right: our output from synthetic test data

HOW THE DOCTOR WORKS TODAY

The process today

1. **Log in** to the online platform and open the case
2. **Open** all the documents individually
3. **Find** diagnoses, date and who issued it
4. **Copy** everything into a Word list
5. **Check** compare the list with the originals

5–15 documents per case, sometimes **up to 80**

By hand every step, for every case

HOW THE DOCTOR WORKS TODAY

What we want to automate

1 **Log in** to the online platform and open the case

→

2 **Open** all the documents individually

→

3 **Find** diagnoses, date and who issued it

→

4 **Copy** everything into a Word list

→

5 **Check** compare the list with the originals

These three steps get automated

5–15 documents per case, sometimes **up to 80**

By hand every step, for every case

BEFORE WE STARTED

The idea and the rules

Runs locally

No cloud. Health data stays on the machine.

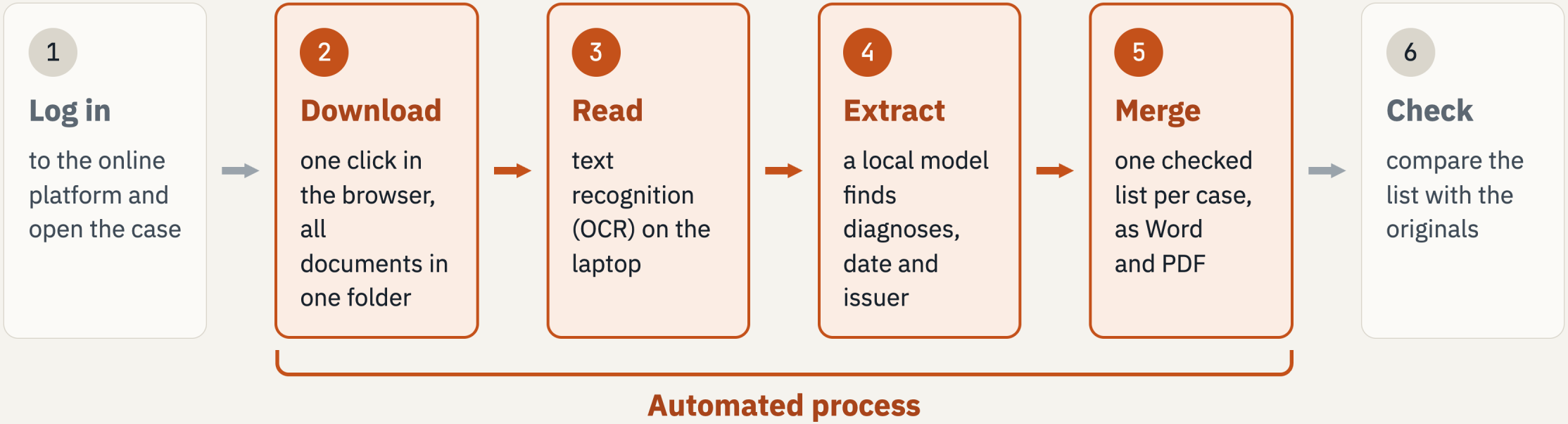
Copy only

Find and label. No interpretation, no medical assessment.

Mark uncertain parts

The source text is never changed silently. Anything uncertain is marked.

The process with the tool



STEP 2 OF 6

02

Download

A small Chrome extension adds a button to the platform’s document list

One click: all medical documents of the case land in one folder

Original files instead of print-outs, so many already have a text layer

The doctor logs in himself, the extension never sees a password

STEP 3 OF 6

03

Reading the text

If the PDF has a text layer: use it directly

If not: the Mac's built-in text recognition (Apple Vision, via the Python package ocrmac)

Runs on the laptop, no service involved

MISTAKES HAPPEN

I69.3 → 169.3

An ICD-10 diagnosis code: the letter I was read as the digit 1. The result is no longer a valid diagnosis code. It is marked in the list, never corrected. (Example from a synthetic test document)

STEP 4 OF 6

04

The local model

LM Studio on a MacBook Air M2 with 16 GB. Tried three models, now in use:
Qwen3 4B

Model	7 short test documents	10 real documents
Gemma 4 12B	354 s	much longer, didn't finish testing
Gemma 3 4B	80 s	not measured
Qwen3 4B	78 s	18 min

Test documents: 1–2 pages each, one request per document.

Real documents, today's setup: about 1 minute per short document, 2–4 minutes per hospital letter.

THE HARD PART

Making a small model work

Small context window

On 16 GB I limit it to 8,000 tokens, so long documents are split into several pieces.

Long instructions confuse it

With all rules in one prompt it sometimes found no diagnoses at all. Every piece now gets two shorter requests.

Two requests

Split into two requests to cover everything.

Inventing

Invented a date when none was readable.

What helped noticeably: short examples in the prompt. **But:** you can’t give an example for every kind of document, there would be far too many.

STEP 5 OF 6

05

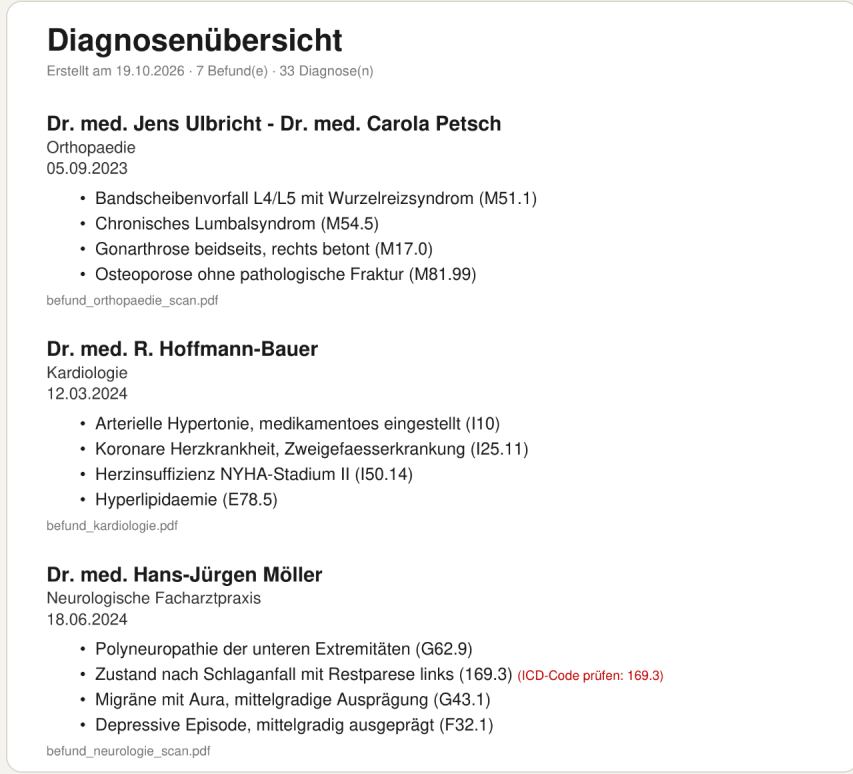
Merging and checking

A plain Python script

Every line the model returns is compared with the text of the document

Not found: printed in red

One list per case, sorted by date, as Word and PDF



Diagnosenübersicht

Erstellt am 19.10.2026 · 7 Befund(e) · 33 Diagnose(n)

Dr. med. Jens Ulbricht - Dr. med. Carola Petsch
Orthopaedie
05.09.2023

- Bandscheibenvorfall L4/L5 mit Wurzelreizsyndrom (M51.1)
- Chronisches Lumbalsyndrom (M54.5)
- Gonarthrose beidseits, rechts betont (M17.0)
- Osteoporose ohne pathologische Fraktur (M81.99)

befund_orthopaedie_scan.pdf

Dr. med. R. Hoffmann-Bauer
Kardiologie
12.03.2024

- Arterielle Hypertonie, medikamentoes eingestellt (I10)
- Koronare Herzkrankheit, Zweigefaesserkrankung (I25.11)
- Herzinsuffizienz NYHA-Stadium II (I50.14)
- Hyperlipidaemie (E78.5)

befund_kardiologie.pdf

Dr. med. Hans-Jürgen Möller
Neurologische Facharztpraxis
18.06.2024

- Polyneuropathie der unteren Extremitäten (G62.9)
- Zustand nach Schlaganfall mit Restparese links (169.3) (ICD-Code prüfen: 169.3)
- Migräne mit Aura, mittelgradige Ausprägung (G43.1)
- Depressive Episode, mittelgradig ausgeprägt (F32.1)

befund_neurologie_scan.pdf

Generated list from synthetic test reports

STATUS TODAY

Where it is stuck: reliability and time

Reliability

- ⚠ **Confused** by long instructions
- ⚠ **Invents** details, e.g. a date
- ⚠ **No room** for an example per document type
- ⚠ **Too small** for the variety of real documents

Only prompt examples, no fine-tuning.

Time

2–4 min per hospital letter

15–25 min per typical case with ten documents, and cases can have up to 80

Not in daily use yet.

Next idea: hardware with 32 or 48 GB and a bigger model, not tested yet.